GNBF5020

期中考試操練手冊

Midterm Drill Pack

20 + 10 分制期中考試完整操練：考試形式、他點名的五種題型、70 條單選最佳答案題庫（含干擾項分析），以及兩份情境題範本。

GNBF5020 · Introduction to Molecular Biology and Genetics
黃教授 Dr. Tony Wong · 第一講至第五講 · 2026 年 9 月 12 日 – 10 月 17 日

> **[ENGLISH ONLY]** The exam must be answered in English. Dr. Wong said this three separate times, and once he said he gave zero marks for it. This document is in English so you can revise directly in the language you will be graded in. The Chinese companion exists only for explanation; gene, protein, pathway, method and drug names stay in English in both versions.

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GNBF5020 —— 期中考試操練手冊

20 + 10 分制期中考試完整操練

GNBF5020 Introduction to Molecular Biology and Genetics · 黃教授 Dr. Tony Wong · 第一講至第五講，2026 年 9 月 12 日 – 10 月 17 日 形式：20 條選擇題 + 短答題三選二 = 30 分 · 90 分鐘 · 只准英文作答

> ***本手冊怎樣用。*** *溫習包裡其他文件教你「內容」；這本操練手冊訓練你「考試形式」：卷面樣貌、他親口點名會出的題型，以及一組可在限時下操練的單選最佳答案題庫。請按次序做。*

1. 考試形式，完全按教授自己的說法

項目	規格	出處
期中比重	**佔總成績 30%**	第一講，Slide 2
期末	佔總成績 40%	第一講，Slide 2
兩份 assignment	佔 30%（「超簡單」）	第一講，Slide 2
A 部	**20 條選擇題，單選最佳答案**	第五講（較近的說法）
B 部	**三條短答題選兩題**（數據分析 / 情境題）	第五講
總分	**30 分，90 分鐘**	第五講
語言	**只准英文作答 —— 用中文答零分**	第一、四、五講

兩種說法不一致 —— 請以較新的為準

講次	日期	教授的說法
第四講	2026-10-10	「about 25 questions, one mark each … two questions, you choose only one short question to answer」
第五講	**2026-10-17**	**「we have 20 multiple choice questions, and two out of three short question … 20 marks … short question is account for 10 marks … 1.5 hours」**

第五講是較近的一次，而且是他自己寫在標題頁上的，所以**請按 20 條選擇題 + 三選二短答題準備**。不論哪個版本，比例與策略完全相同。

> ***他自己講的策略：*** *「multiple choice is for you to pass the exam. But if you want to get a full mark or a better result, you'd better perform well for the short questions, which emphasize on data analysis and also the scenarios questions.」—— 第一講*

即是說：**20 條選擇題是下限；短答題那 10 分才是分數的差距所在。**

90 分鐘時間分配

分鐘	做法	原因
0–5	讀完整份卷；決定要作答哪兩條短答題	三選二由你決定，要早決定
5–35	完成 20 條選擇題，每題約 90 秒	快分；不要糾結
35–80	作答已選的兩條短答題，每條約 22 分鐘	分數主要在這裡
80–88	重看不確定的選擇題	重讀一遍是便宜的分數
88–90	檢查有沒有寫完整句子、有沒有用英文	這一項可以拿零分

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2. 短答題的四條規則

以下來自他在各講的說法，每一條以往都有人因此失分。

1. **一律用英文作答。** 他講過三次，而且說過去年他**直接給零分**。簡單的英文句子也比完全沒有英文好。
2. **寫完整句子。** 他的建議是 *defend your reasoning*（為你的推理辯護）。**一整頁 bullet points 不算答案。** bullet point 是筆記；答案是一段有論證的句子。
3. **句子裡面的技術名詞保留英文。** 請寫「the KRAS mutation means cetuximab will not work」，而不是一句中文裡面夾一個英文詞。
4. **寫出推理鏈，不只寫結論。** 每條情境題都要標步驟。有鏈但沒結論得部分分；有鏈也有結論才得滿分。

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3. 他點名為考題的五種題型

這是全科最有把握的題目。他親口說過，這些就是他放進考卷的題型。**要學會每題的那條推理鏈，不要只背答案。**

3.1 讀 Kaplan–Meier 曲線 —— 第一講

> *「And I ask this single question every, every time when I talk about the KM plot … And only a single student give me a good answer last year.」—— 第一講*

座標軸的意思 - **x 軸：** 時間（月） - **y 軸：** 存活機率，或無進展存活機率 - 曲線**由 1.0 開始**，以**階梯狀**下降

每個元素怎樣讀

元素	意思
平的一段	該區間沒有事件發生
向下的一級	發生了一次事件（死亡或進展）
曲線上的小刻痕	**Censored（截尾）**病人 —— 失訪，或到截止點時仍然存活
早期陡降	該基因與**早期 involvement** 有關，預後較差
後期才下降 / 長平台	病人早期表現良好；事件與**晚期 involvement** 有關
中位存活期	曲線穿越 0.5 的位置

答案要畫的推理鏈

1. 先讀座標軸　→ x 是什麼、y 是什麼、單位是什麼
2. 認出各組　→ 哪條線是哪個治療 / 哪個基因型
3. 找分離點　→ 曲線在哪裡分開？分開多少？
4. 判斷時間意義　→ 早期分離 = 早期 involvement；晚期 = 晚期 involvement
5. 說出臨床用途　→ 這個數據支持哪一個 biomarker 決策？
6. 加上限制說明　→ 有 censored 刻痕，代表估計偏保守

要寫的句子：「The curve for group A falls steeply within the first six months, whereas group B has a long plateau before any step, so the gene under investigation is associated with early involvement and a poorer prognosis in group A.」

3.2 從醫療報告做遺傳諮詢 —— 第一講

> *「if you're a genetic consultant and I give you a medical report, how you would suggest the client whether she or him will get a baby with a certain genetic diseases?」—— 第一講*

答案要畫的推理鏈

1. 先由家系圖判斷遺傳方式
   - autosomal dominant → 連代出現、男女皆病、可有父傳子
   - autosomal recessive → 隔代出現、兄弟姊妹患病而父母正常
   - X-linked recessive → 多為男性、無父傳子
   - mitochondrial → 患病母親傳給全部子女；患病父親一個也不傳
2. 先確定遺傳方式，再算風險
3. 然後算這一對配偶的風險
   - autosomal recessive：雙方帶因機率 × 每次懷孕 1/4
   - X-linked recessive：視乎母親的帶因狀態
   - mitochondrial：母親患病則所有子女都會有；父親無關
4. 說明是「每次懷孕」的機率（不是每個孩子的機率）
5. 說出下一步 —— 遺傳檢測、或做 IVF 配合胚胎植入前檢測

要寫的句子：「The pedigree is most consistent with autosomal recessive inheritance, because affected siblings are born to two unaffected parents and the trait skips a generation. If both parents are obligate carriers, the risk to each pregnancy is one in four. I would recommend testing both parents to confirm carrier status, and, if they wish to avoid transmission, preimplantation genetic testing with IVF.」

3.3 NHEJ 缺陷導致 SCID —— 第四講

> *「if I ask you, if the people showing a defective in this non-homologous end joining, what's the outcome? Definitely the outcome is the severe immune deficiency … because he cannot generate a sufficient amount of the antibody pool.」—— 第四講*

答案要畫的推理鏈

NHEJ → V(D)J recombination → 把 V、D、J 片段接合起來
→ 產生 B 細胞的抗體多樣性 + T 細胞的受體多樣性
→ 多樣性庫根本建不起來
→ SCID（嚴重聯合性免疫缺陷）

他給的數字： V ≈ 39 份、D = 27 份、J = 6 份。

要寫的句子：「NHEJ is co-opted by V(D)J recombination to join the V, D and J gene segments, and this is what generates antibody diversity in B-cells and T-cell receptor diversity in T-cells. A defect in NHEJ therefore prevents the repertoire from being built at all, and the clinical outcome is severe combined immunodeficiency.」

3.4 讀粒線體家系圖 —— 第二講

> *「Not only the mutation within the mitochondrial DNA can cause the mitochondrial disease. Some mitochondrial disease can actually result from the mutation within the nuclear DNA … different mutation can cause the different inheritance patterns.」 —— 第二講*

粒線體（母系）遺傳的三個特徵 1. **患病母親**把疾病傳給**全部**子女 2. **患病父親一個也不傳** 3. **只有女性**可以繼續傳下去

陷阱： 疾病本身也可能是**核 DNA** 突變造成，而那樣就會**按孟德爾方式遺傳**。不要只憑家系圖就認定是粒線體遺傳 —— 要問**是哪個基因**。

答案要畫的推理鏈

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1. 先看是誰傳給誰、誰沒有傳
2. 母系型？ → mitochondrial DNA 突變
3. 孟德爾型？ → 核 DNA 突變（所以不是粒線體遺傳）
4. 說出基因 / 類型
5. 給出再發風險
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要寫的句子：「The pedigree shows transmission through affected mothers to all their children and none from affected fathers, which is the maternal pattern of mitochondrial inheritance and therefore points to a mutation in mitochondrial DNA. Because the condition can also be caused by nuclear-gene mutations, I would confirm the genotype before giving the final recurrence risk.」

3.5 把 lac operon 的邏輯讀對 —— 第五講

> *Tier 2 —— 他說你必須理解。案例研習題庫的案例 16 也是這一題。*

要強烈轉錄必須同時成立的兩個條件

條件	作用	分子
有 lactose	Allolactose 結合 Lac repressor，把它從 operator 推開	**Allolactose**（由 lacZ 編碼的 β-galactosidase 製造 —— **不是** lactose 本身）
glucose 低	cAMP 上升；cAMP–CAP 複合體結合 promoter，把複製機器招來	**cAMP–CAP**

> ***只答「lactose」是經典錯答。*** *最好也只得一半分。*

三個基因： **lacZ**（β-galactosidase）、**lacY**（permease）、**lacA**（transacetylase）—— 一起轉錄成一條 polycistronic message。

trp operon 是完全相反的邏輯： tryptophan 作為 **corepressor** 結合 Trp repressor，關掉 tryptophan 的生合成。它的五個基因：**trpE, trpD, trpC, trpB, trpA**。

IPTG 是重組 *E. coli* 系統使用的合成誘導物，不會被代謝。他投影片上的最佳濃度：**100 μM**（測試範圍 0–1000 μM）。

要寫的句子：「Strong transcription of the lac operon needs two conditions at once: lactose must be present so that allolactose binds the Lac repressor and removes it from the operator, and glucose must be low so that cAMP rises and the cAMP-CAP complex binds the promoter. Either condition alone gives only weak transcription.」

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4. 選擇題操練庫 —— 60 條單選最佳答案題

請在限時下操練：**每題 90 秒**。如果答不出來，先標記再跳過；你有 20 分的下限要保住。

每題答案都列出**最容易令人失分的干擾項**，因為他講過選擇題失分的原因正是「try to select multiple options」。

> *下列題目、選項與答案全部保留英文，因為考試就是用英文作答；中文只提供解釋。*

第一講 —— Introduction to Biomolecules

Q1. Which statement about biomolecules is correct? - A. All four classes are polymers built from monomers - B. Lipids are polymers of fatty acids joined by peptide bonds - C. Carbohydrates, lipids, proteins and nucleic acids are the four classes of biomolecule - D. Proteins are the only class that form by dehydration condensation - E. Nucleic acids are built from amino acids

▶ **Answer**

Q2. Which bond links the two nucleotides of a phosphodiester linkage? - A. A hydrogen bond between complementary bases - B. A peptide bond between the 3′-OH and the 5′-phosphate - C. A glycosidic bond between two sugars - D. A disulfide bridge between two cysteines - E. A π bond within the base

▶ **Answer**

Q3. Why does DNA contain thymine rather than uracil? - A. Thymine pairs with guanine - B. Uracil is unstable at physiological pH - C. Deamination of cytosine produces uracil, which is detectable and repairable; deamination of 5-methylcytosine produces thymine, which is not - D. Thymine has three hydrogen bonds - E. RNA is single-stranded and cannot contain uracil

▶ **Answer**

Q4. The primary level of protein structure is: - A. The α-helix and β-sheet - B. The sequence of amino acids joined by peptide bonds - C. The three-dimensional fold of the whole chain - D. The association of several subunits - E. Covalent bonds between distant residues

▶ **Answer**

Q5. A molecule that changes shape — and therefore its activity — when something binds at a site other than its active site is showing: - A. Denaturation - B. Allostery - C. Hydrolysis - D. Isomerisation - E. Saturating kinetics

▶ **Answer**

Q6. Which of these is *not* a function of lipids? - A. Energy storage - B. Membrane structure - C. Insulation - D. Catalysis - E. Signalling

▶ **Answer**

Q7. Alpha-helices and beta-sheets are examples of: - A. Primary structure - B. Secondary structure - C. Tertiary structure - D. Quaternary structure - E. Nucleotide sequence

▶ **Answer**

Q8. The bond that stabilises the tertiary structure of a protein by linking two cysteine residues is: - A. An ionic bond - B. A hydrogen bond - C. A peptide bond - D. A disulfide bond - E. A phosphodiester bond

▶ **Answer**

Q9. Which of the following is a monosaccharide? - A. Cellulose - B. Starch - C. Glucose - D. Sucrose - E. A nucleotide

▶ **Answer**

Q10. In a nucleotide, the nitrogenous base is attached to: - A. The 1′ carbon of the sugar - B. The 3′ carbon of the sugar - C. The 5′ carbon of the sugar - D. The phosphate group - E. The 3′ hydroxyl of the adjacent nucleotide

▶ **Answer**

第二講 —— Basic Cell Biology

Q11. Which structure is *not* found in a prokaryotic cell? - A. Ribosomes - B. A plasma membrane - C. A mitochondrion - D. Circular DNA - E. A nucleoid region

▶ **Answer**

Q12. A red blood cell placed in pure distilled water will: - A. Remain unchanged - B. Shrink - C. Burst (lyse) - D. Become crenated - E. Lose its nucleus only

▶ **Answer**

Q13. A red blood cell placed in a hypertonic solution becomes: - A. Hemolysed - B. Crenated (spiky and shrunken) - C. Larger and spherical - D. Converted to a reticulocyte - E. Unaffected

▶ **Answer**

Q14. Transport across the membrane that requires ATP is: - A. Simple diffusion - B. Facilitated diffusion - C. Osmosis - D. Active transport - E. Filtration

▶ **Answer**

Q15. The rough endoplasmic reticulum is studded with ribosomes because it is the site of: - A. Lipid synthesis - B. Protein synthesis for secretion and membranes - C. ATP production - D. DNA replication - E. Lysosomal degradation

▶ **Answer**

Q16. Mitochondria and chloroplasts are thought to have originated from: - A. Free-living bacteria engulfed by an ancestral cell - B. Folding of the plasma membrane - C. Division of the Golgi apparatus - D. Invagination of the nuclear envelope - E. Endocytosis of a large vacuole

▶ **Answer**

Q17. In a pedigree, an affected mother passes the trait to all her children and an affected father passes it to none. The inheritance is: - A. Autosomal dominant - B. Autosomal recessive - C. X-linked recessive - D. Mitochondrial - E. Y-linked

▶ **Answer**

Q18. Which phase of the cell cycle is DNA replicated? - A. G1 - B. S - C. G2 - D. M - E. Cytokinesis

▶ **Answer**

Q19. During mitosis, the sister chromatids separate and move to opposite poles in: - A. Prophase - B. Metaphase - C. Anaphase - D. Telophase - E. Prometaphase

▶ **Answer**

Q20. Crossing over occurs during: - A. Prophase I of meiosis - B. Metaphase I of meiosis - C. Anaphase I of meiosis - D. Mitosis - E. S phase

▶ **Answer**

第三講 —— DNA and Chromatin Structure

Q21. The two strands of a DNA double helix are: - A. Complementary and antiparallel - B. Complementary and parallel - C. Identical and antiparallel - D. Identical and parallel - E. Non-complementary and antiparallel

▶ **Answer**

Q22. Which statement about the major groove is correct? - A. It is wider than the minor groove - B. It is wider than the minor groove and is where many proteins read DNA sequence - C. It contains the sugar-phosphate backbone - D. It has no hydrogen bonds - E. It is found only in RNA

▶ **Answer**

Q23. A nucleosome consists of: - A. DNA wrapped around a histone octamer - B. RNA wrapped around a polymerase - C. A histone octamer wrapped around RNA - D. Free DNA with no proteins - E. A single H1 histone with 146 bp of DNA

▶ **Answer**

Q24. Which histone modification is most associated with **active** transcription? - A. H3K9me3 - B. H3K27me3 - C. H3K4me3 - D. H3K9me2 - E. DNA methylation at CpG islands

▶ **Answer**

Q25. Heterochromatin differs from euchromatin in that it is: - A. Less condensed and more transcriptionally active - B. More condensed and less transcriptionally active - C. Identical, but found only in the nucleus - D. Enriched in acetylated histones - E. The form found only during S phase

▶ **Answer**

Q26. The 5′ end of a eukaryotic linear chromosome is protected by: - A. A centromere - B. A telomere - C. An origin of replication - D. A satellite DNA block - E. A nucleosome

▶ **Answer**

Q27. The centromere of a chromosome is the region where: - A. Replication begins - B. Telomerase binds - C. Spindle microtubules attach via the kinetochore - D. RNA polymerase transcribes - E. Histone H1 is removed

▶ **Answer**

Q28. Which pair of bases forms **three** hydrogen bonds? - A. A–T - B. G–C - C. A–G - D. U–A - E. T–G

▶ **Answer**

Q29. In which cellular compartment is DNA replication carried out in eukaryotes? - A. The cytoplasm - B. The nucleus, during S phase - C. The mitochondrion only - D. The rough ER - E. The nucleolus

▶ **Answer**

Q30. Which of these is **not** a level of DNA packaging in eukaryotes? - A. Nucleosome - B. 30-nm fibre - C. Looped domains attached to a scaffold - D. Chromatid - E. Telomere

▶ **Answer**

Q31. Base stacking contributes to DNA stability mainly through: - A. Covalent bonding of the backbone - B. Hydrophobic interactions between adjacent base pairs - C. Ionic bonds with magnesium - D. Phosphodiester linkages - E. Disulfide bridges

▶ **Answer**

Q32. A “purine” is: - A. A base with one ring - B. A base with two fused rings - C. Any base that pairs with two others - D. A sugar-phosphate unit - E. A type of histone

▶ **Answer**

第四講 —— DNA Replication, Repair and Recombination

Q33. Meselson and Stahl's result supports: - A. Conservative replication - B. Dispersive replication - C. Semi-conservative replication - D. Semi-discontinuous replication only - E. Rolling-circle replication

▶ **Answer**

Q34. Which enzyme relieves the torsional stress that builds up ahead of the replication fork? - A. Helicase - B. Topoisomerase - C. Primase - D. Ligase - E. Telomerase

▶ **Answer**

Q35. Why can DNA polymerase not start a strand from nothing? - A. It lacks the energy to open the double helix - B. It can only add nucleotides to an existing 3′-OH, so a primer is required - C. It cannot bind the template without ATP - D. It is inhibited by single-strand binding proteins - E. It can only add to the 5′ end

▶ **Answer**

Q36. On the lagging strand, the discontinuously made fragments are called: - A. Okazaki fragments - B. DnaA boxes - C. Telomeres - D. Centromeres - E. Exons

▶ **Answer**

Q37. Telomerase is: - A. A DNA polymerase that copies the lagging strand - B. A reverse transcriptase that uses its own RNA as a template - C. A helicase - D. A repair endonuclease - E. A topoisomerase

▶ **Answer**

Q38. Telomerase activity is found in stem cells and germ cells, and in: - A. Most somatic cells - B. About 90% of cancer cells - C. No human cells - D. Only mature red blood cells - E. Neurons only

▶ **Answer**

Q39. Which pathway removes a bulky lesion such as a UV-induced pyrimidine dimer? - A. Base excision repair - B. Nucleotide excision repair - C. Mismatch repair - D. Direct reversal by photolyase - E. Homologous recombination

▶ **Answer**

Q40. Which pathway fixes replication errors that escaped polymerase proofreading? - A. Base excision repair - B. Nucleotide excision repair - C. Mismatch repair - D. Fanconi anaemia pathway - E. Apoptosis

▶ **Answer**

Q41. A defect in nucleotide excision repair causes: - A. Xeroderma pigmentosum - B. Lynch syndrome - C. Hereditary breast and ovarian cancer - D. Cystic fibrosis - E. Sickle cell anaemia

▶ **Answer**

Q42. The genes BRCA1 and BRCA2 function in: - A. Mismatch repair - B. Base excision repair - C. Homologous recombination - D. Nucleotide excision repair - E. Non-homologous end joining

▶ **Answer**

Q43. Which statement about non-homologous end joining is correct? - A. It requires a sister chromatid as a template and is accurate - B. It joins broken ends without a template and may lose or gain bases - C. It excises a patch of nucleotides around the lesion - D. It requires photolyase - E. It only functions during S phase

▶ **Answer**

Q44. A defect in NHEJ causes severe combined immunodeficiency because the cell cannot: - A. Synthesize DNA - B. Perform V(D)J recombination and so build antibody and T-cell receptor diversity - C. Replicate the mitochondrial genome - D. Perform base excision repair - E. Translate mRNA

▶ **Answer**

Q45. Homologous recombination during meiosis produces: - A. Sister chromatids - B. Non-sister recombinant chromatids - C. Identical daughter cells - D. Polyploid cells - E. Acrocentric chromosomes

▶ **Answer**

Q46. A transposon is best described as: - A. A protein that moves a chromosome to the nucleus - B. A mobile DNA element that moves its position within the genome - C. A type of spindle microtubule - D. A repair enzyme that removes thymine dimers - E. A region of a chromosome where the centromere forms

▶ **Answer**

Q47. Which of these mutagens is an alkylating agent that adds an ethyl group to a DNA base and commonly produces G→A transitions? - A. Ethidium bromide - B. Ethyl methanesulphonate (EMS) - C. Ultraviolet light - D. 5-bromouracil - E. Aflatoxin B1

▶ **Answer**

Q48. Which of these is **not** a spontaneous (endogenous) source of DNA damage? - A. Tautomerisation of bases - B. Depurination - C. Deamination of cytosine to uracil - D. Ultraviolet radiation - E. Oxidative damage by reactive oxygen species

▶ **Answer**

Q49. Direct reversal repair of a UV-induced pyrimidine dimer is carried out by: - A. Photolyase, using visible light energy - B. UvrABC, using ATP - C. DNA polymerase III - D. Ligase - E. Helicase II

▶ **Answer**

Q50. A cell with a defective DNA ligase would most obviously fail to: - A. Unwind the helix - B. Join Okazaki fragments on the lagging strand - C. Synthesize RNA primers - D. Remove uracil from DNA - E. Replicate the mitochondrial genome

▶ **Answer**

Q51. Which of these best distinguishes homologous recombination from NHEJ? - A. HR happens only in bacteria - B. HR uses a homologous template; NHEJ does not - C. NHEJ requires ATP and HR does not - D. HR occurs only during S phase - E. NHEJ is the only pathway that can repair double-strand breaks

▶ **Answer**

Q52. A thymine dimer distorts the DNA helix. Which pathway is designed to remove it? - A. Mismatch repair - B. Base excision repair - C. Nucleotide excision repair - D. Direct reversal by MGMT - E. Fanconi anaemia pathway

▶ **Answer**

第五講 —— Control of Gene Expression

Q53. The lac operon is switched on *strongly* only when: - A. Lactose is present - B. Glucose is low - C. Both lactose is present and glucose is low - D. Both lactose is absent and glucose is high - E. Tryptophan is scarce

▶ **Answer**

Q54. The molecule that inactivates the Lac repressor is: - A. Lactose itself - B. Glucose - C. Allolactose - D. cAMP - E. IPTG only in wild type

▶ **Answer**

Q55. The cAMP–CAP complex helps RNA polymerase by: - A. Binding the operator and blocking transcription - B. Binding the promoter and recruiting the transcription machinery - C. Degrading the operator - D. Converting glucose into lactose - E. Methylating the promoter

▶ **Answer**

Q56. The three structural genes of the lac operon are: - A. lacZ, lacY, lacA - B. trpE, trpD, trpC - C. araA, araB, araC - D. cAMP, CAP, and the repressor - E. p53, Rb and APC

▶ **Answer**

Q57. In the trp operon, tryptophan acts as: - A. An inducer that activates transcription - B. A corepressor that activates the repressor - C. An allosteric activator of CAP - D. A substrate of β-galactosidase - E. A transcriptional activator

▶ **Answer**

Q58. The five genes of the tryptophan operon are: - A. lacZ, lacY, lacA, lacI, lacO - B. trpE, trpD, trpC, trpB, trpA - C. V, D, J, RAG1, RAG2 - D. TP53, RB1, APC, KRAS, BRAF - E. p16, p21, p27, p53, Rb

▶ **Answer**

Q59. A **cis**-acting element is: - A. A diffusible protein that binds a distant gene - B. DNA sequence on the same DNA molecule that regulates the gene next to it - C. A transcription factor made in another cell - D. A microRNA - E. A histone mark

▶ **Answer**

Q60. The clearest example of trans-regulation is: - A. An enhancer on chromosome 13 activating a gene on chromosome 13 - B. A repressor encoded on chromosome 1 acting on a target gene on chromosome 13 - C. A promoter on the same chromosome as its gene - D. An operator next to the structural gene - E. A poly-A signal in the 3′ UTR

▶ **Answer**

Q61. Alternative splicing results in: - A. One gene producing several different proteins - B. Several genes producing one protein - C. Loss of all introns - D. Failure of transcription - E. Hyper-methylation of the promoter

▶ **Answer**

Q62. An RNA longer than 200 nucleotides that makes no protein is called: - A. A microRNA - B. A long non-coding RNA (lncRNA) - C. A small nuclear RNA - D. A tRNA - E. An siRNA

▶ **Answer**

Q63. A microRNA works by: - A. Binding a seed sequence in the target mRNA by partial base-pairing to suppress its translation - B. Binding the promoter and blocking RNA polymerase - C. Methylating the target gene - D. Cutting the DNA of the target gene - E. Recruiting histone deacetylase to the promoter only

▶ **Answer**

Q64. In bisulfite sequencing, unmethylated cytosine is: - A. Left unchanged - B. Converted to uracil, while methylated cytosine is left alone - C. Converted to thymine, while methylated cytosine is left alone - D. Converted to 5-methylcytosine - E. Degraded completely

▶ **Answer**

Q65. Which of these is a *trans*-acting epigenetic mechanism? - A. Promoter CpG methylation - B. Histone acetylation - C. Histone deacetylation - D. A microRNA - E. DNA methylation

▶ **Answer**

Q66. Which statement about cellular memory is correct? - A. A cell keeps a switch on long after the original signal has gone - B. A cell permanently loses the ability to change - C. It requires continuous signalling from outside the cell - D. It only occurs in prokaryotes - E. It is caused by DNA replication

▶ **Answer**

Q67. Which of these would be the expected effect of **hypomethylation** of a tumour suppressor promoter? - A. Increased expression of the tumour suppressor - B. Silencing of the tumour suppressor - C. No change in transcription - D. Increased RNA editing - E. Enhanced mismatch repair

▶ **Answer**

Q68. In CHO cells used for producing a therapeutic protein, Dr. Wong said you must: - A. Introduce a nuclear localisation signal - B. Perform codon optimisation - C. Use a prokaryotic vector - D. Add a poly-A tail to the genomic

DNA - E. Delete the introns

▶ **Answer**

Q69. IPTG is used in recombinant *E. coli* systems because it: - A. Is a sugar that the cell can metabolise for energy - B. Is an artificial inducer that is not metabolised, so it switches on the lac system without being consumed - C. Directly binds RNA polymerase - D. Degrades lactose - E. Acts as a corepressor for the trp operon

▶ **Answer**

Q70. The NF-κB cytokine response proceeds in the order: - A. IFN-γ → IL-1β → TNF-α → IL-2 - B. TNF-α → IL-1β → IFN-γ → IL-2 and IL-15 - C. IL-2 → TNF-α → IFN-γ → IL-1β - D. TNF-α → IL-2 → IL-1β → IFN-γ - E. All four appear simultaneously

▶ **Answer**

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5. 兩份情境題範本

他點名的短答題只有這兩大類。**把這兩類練熟就好，不必平均用力。**

範本 A —— 家系圖 / 遺傳諮詢

第一步：先描述家系圖的型態，再命名。 數每一代有幾位患病、患病者的性別、檢查有沒有父傳子。

你看到的型態	符合
連代都有、男女皆病、可父傳子	Autosomal dominant
隔代出現、兄弟姊妹患病而父母正常	Autosomal recessive
多為男性、無父傳子	X-linked recessive
患病母親 → 全部子女；患病父親 → 全部不傳	Mitochondrial
患病父親 → 只傳給全部兒子	Y-linked

第二步：算這一對配偶的風險。 - Autosomal recessive，雙方皆為帶因者：**每次懷孕 1/4** - Autosomal dominant，heterozygous 的一方：**每次懷孕 1/2** - X-linked recessive，母親為帶因者：**兒子一半患病，女兒一半帶因** - Mitochondrial，母親患病：**患病母親所有子女都會有；患病父親的子女一個也不會有**

第三步：說出下一步。 用 genetic testing 確認；如希望避免傳遞，可考慮 IVF 配合 preimplantation genetic testing；並就疾病本身提供諮詢。

第四步：用完整句子、用英文作答，基因與疾病名稱保留英文。

範本 B —— 存活數據 / Kaplan–Meier

第一步：讀座標軸。 x = 時間（月）；y = 存活機率或無進展存活機率。曲線由 1.0 開始。

第二步：認出各組。 哪條線是哪個治療、哪個基因型、哪個 biomarker 狀態？

第三步：找分離點與其時間。 - 曲線直到某個月才分開 → 效應出現得晚 - 一開始就分開 → 效應一早就存在 - 早期陡降的那組 → 預後較差

第四步：讀數字。 中位存活期（曲線穿越 0.5 處）；每組人數；如有 log-rank p-value 一併指出。

第五步：連到臨床決策。 你會驗哪一個 biomarker？會選哪一種治療？這條曲線支持 predictive 還是 prognostic 的說法？

第六步：說出限制。 刻痕代表 censored 病人 —— 失訪，或到截止點時仍然無事件。Censoring 令估計偏保守，真實曲線可能更差。

第七步：用完整句子、用英文作答。

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6. 常見失分陷阱

陷阱	為何失分	應該怎樣做
用中文作答	零分 —— 他說去年真的這樣做	用英文寫，即使句子簡單
只有 bullet points	「一整頁 bullet points 不算答案」	寫有論證的完整句子
lac operon 只答 lactose	漏了第二個條件	一定寫「lactose 存在 **而且** glucose 低」
以為所有粒線體病都是粒線體遺傳	也可能是核基因突變	先看家系圖，再問是哪個基因
把 UVA 損傷說成 BER	BER 是 base excision	bulky lesion 用 NER，單一鹼基用 BER
選擇題寫「A 和 B」	單選最佳答案	選一個；用排除法
一條選擇題花 5 分鐘	你有 20 分的下限要守	每題 90 秒，標記後跳過
忘記寫時間單位	沒有單位的曲線沒有意義	一定先講 x 軸與 y 軸的單位
不寫「每次懷孕」	「50% 風險」有歧義	寫明 per pregnancy
引用 lecturer 說法時自行改寫	失去可信度	原句照抄英文

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7. 考前七天計劃

日	任務	時間
1	讀 §1–§2（考試形式、四條寫作規則）。把形式寫在一張卡上。	30 分鐘
2	熟記 **lac operon** 邏輯（§3.5）與 **NHEJ → SCID** 推理鏈（§3.3），要能出聲講出來	45 分鐘
3	限時操練選擇題 Q1–Q24（每題 90 秒）。每一條錯題都做記號。	45 分鐘
4	限時操練選擇題 Q25–Q48。**不看答案**重做所有錯題。	45 分鐘
5	操練選擇題 Q49–Q70。然後重做所有錯題。	45 分鐘
6	練 §5 兩份範本：用英文完整句子、在 22 分鐘內寫出兩條短答題。	60 分鐘
7	重讀 §3.1（Kaplan–Meier）、§3.2（遺傳諮詢）、§3.4（粒線體家系	30 分鐘

日	任務	時間
	圖）。睡 8 小時。	

合計約 5 小時。 只要短答題真的用完整英文句子寫，這 5 小時是夠的。

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8. 最後自查

只有在**今天、不看資料**也能做到，才可以打勾。

- ☐ 我能說出考試形式：20 條選擇題 + 三選二短答題、30 分、90 分鐘、只准英文。
- ☐ 我知道期中佔總成績 30%。
- ☐ 我能用英文完整句子解釋 NHEJ 失效為何導致 SCID。
- ☐ 我能說出 lac operon 強烈轉錄的兩個條件，以及各自由哪個分子負責。
- ☐ 我能讀 Kaplan–Meier 曲線，判斷基因是早期還是晚期 involvement。
- ☐ 我能在家系圖中辨認粒線體遺傳，**並**解釋核基因突變為何可能造成同樣外觀。
- ☐ 我能由家系圖算出每次懷孕的風險，並說出建議做哪項檢測。
- ☐ 我能按「各修哪一種損傷」區分 BER、NER、MMR、HR、NHEJ。
- ☐ 我至少用完整英文句子寫過兩條短答題。
- ☐ 我能說出 trp operon 的五個基因與 lac operon 的三個基因。
- ☐ 我知道 IPTG 最佳濃度，以及 CHO 細胞為何需要 codon optimisation。
- ☐ 我分得清 cis element 與 trans factor，並記得 chromosome 1 / chromosome 13 的例子。

勾滿 10 條以上才算準備好。少於 10 條，就把未做過的那幾天的內容補回來。

以上所有題目都必須用英文作答。寫完整句子。為你的推理辯護。

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